1. Define Programming: What is programming, and what is its primary purpose in today's technological world?

Programming is the process of creating a set of instructions that tells a computer how to perform a specific task. Its primary purpose is to solve problems, automate repetitive tasks, and facilitate human-computer interaction, powering everything from smartphone apps to complex scientific simulations.

2. Algorithm Characteristics: List and briefly explain at least three essential characteristics of a good algorithm.

- Finiteness: An algorithm must always terminate after a finite number of steps.
- Definiteness: Each step of the algorithm must be precisely defined; the actions to be carried out must be rigorously and unambiguously specified.
- Input/Output: An algorithm should have zero or more well-defined inputs and at least one well-defined output.

3. Flowchart Symbols: Draw and explain the purpose of at least five standard flowchart symbols.

- Terminator (Oval): Represents the Start or End of a process.
- Process (Rectangle): Represents a specific operation or action (e.g., calculations).
- Decision (Diamond): Represents a branching point where a question is asked (True/False).
- Input/Output (Parallelogram): Represents the act of inputting data or displaying results.
- Connector (Small Circle): Used to connect different parts of a flowchart on the same page.

4. Algorithm for Simple Interest: Write an algorithm to calculate simple interest given principal, rate, and time.

1. Start.
2. Input values for Principal (P), Rate (R), and Time (T).
3. Calculate Simple Interest (SI) using the formula: SI = (P * R * T) / 100.
4. Print the value of SI.
5. Stop.

5. Flowchart for Maximum of Two: Draw a flowchart to find the larger of two given numbers.

6. Algorithm for Celsius to Fahrenheit: Write an algorithm to convert temperature from Celsius to Fahrenheit.

Answer:

1. Start.
2. Input temperature in Celsius (C).

3. Apply formula: F = (C * 9/5) + 32
4. Print temperature in Fahrenheit (F).
5. Stop.

7. Identify Data Types: Suggest the most appropriate data type (General & Java).

Answer:

Data	General Type	Java Type
Your age	Integer	int
Your full name	String	String
Price of a product	Floating-point	double
Light on/off	Boolean	boolean
Number of students	Integer	int
Single letter grade	Character	char

8. Operator Types: Categorize the following operators.

Answer:

+ : Arithmetic

&& : Logical (AND)

> : Relational

= : Assignment

% : Arithmetic (Modulus)

!= : Relational (Not Equal)

|| : Logical (OR)

*= : Compound Assignment

9. Comments: Explain the importance of comments in programming. How do single-line and multi-line comments differ in Java?

Comments improve code readability and help other developers (or your future self) understand the logic.

- Single-line: Starts with //. Everything after this on the same line is ignored.

- Multi-line: Starts with /* and ends with */. It can span several lines.

10. Pseudocode vs. Flowchart: Discuss the advantages and disadvantages.

- Flowchart: High visual clarity and easy to follow logic flow, but difficult to modify and time-consuming to draw for large programs.
- Pseudocode: Faster to write and easily translated into actual code, but lacks the visual impact and can be harder for non-programmers to grasp at a glance.

Java Coding Exercises

11. Hello World

```
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, Java World!");
    }
}
```

12. Variable Declaration & Initialization

```
int myNumber = 25;
double piValue = 3.14159;
boolean isJavaFun = true;
char myInitial = 'G'; // Replace with your initial

System.out.println(myNumber);
System.out.println(piValue);
System.out.println(isJavaFun);
System.out.println(myInitial);
```

13. Arithmetic Operations

```
double num1 = 15.0, num2 = 4.0;
System.out.println("Sum: " + (num1 + num2));
System.out.println("Diff: " + (num1 - num2));
System.out.println("Prod: " + (num1 * num2));
System.out.println("Div: " + (num1 / num2));
System.out.println("Mod: " + (num1 % num2));
```

14. User Input - Sum of Two Numbers

```
import java.util.Scanner;
```

```java
Scanner sc = new Scanner(System.in);
System.out.print("Enter two numbers: ");
int n1 = sc.nextInt();
int n2 = sc.nextInt();
System.out.println("Sum: " + (n1 + n2));
```

15. Area of a Circle

```java
System.out.print("Enter radius: ");
double r = sc.nextDouble();
double area = Math.PI * r * r;
System.out.println("Area: " + area);
```

16. Swap Two Numbers (Without third variable)

```java
int a = 10, b = 20;
a = a + b; // a = 30
b = a - b; // b = 10
a = a - b; // a = 20
```

17. Relational Operators

```java
int a = 10, b = 20;
System.out.println(a == b); // false
System.out.println(a != b); // true
System.out.println(a > b);  // false
```

18. Logical Operators

```java
boolean isSunny = true, isWarm = false;
System.out.println(isSunny && isWarm); // false
System.out.println(isSunny || isWarm); // true
System.out.println(!isSunny);          // false
```

19. Type Casting

```java
int myInt = 100;
double myDouble = 20.5;
int castedInt = (int) myDouble; // Explicit
double result = myInt + myDouble; // Implicit
System.out.println("Casted: " + castedInt + " Result: " + result);
```

20. Increment/Decrement

```
int count = 5;
System.out.println(++count); // 6
System.out.println(count++); // 6 (then becomes 7)
System.out.println(--count); // 6
System.out.println(count--); // 6 (then becomes 5)
```